

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **large-scale distributed systems** supporting enterprise Java Spring Boot microservices across **AWS infrastructure** (EC2, S3, Lambda, CloudWatch), implementing **data structures and algorithms** for high-throughput transaction processing while reducing P1 incident response time by **40%** through **distributed system design patterns** and reliability optimization.
- Designed and implemented **product development code** in **Java Spring Boot** and **Python** for REST API development, feature engineering, and bug fixes on interconnected distributed services; conducted **code reviews** with engineering peers to enforce **best practices** including style guidelines, testability, and efficiency across **3+ concurrent microservices**.
- Architected **compute and storage infrastructure** changes for enterprise applications using **Docker containerization**, **Jenkins CI/CD pipelines**, and **Git version control**; led **design reviews** with stakeholders to evaluate technology trade-offs for distributed system components, achieving **35% faster deployment cycles** and improved system reliability.
- Triaged and resolved **product and system issues** on production **distributed network services** by analyzing root causes across **hardware, network, and service operations** using **Splunk, Dynatrace, and CloudWatch monitoring**; debugged and tracked issues impacting SLA compliance, reducing mean time to resolution (MTTR) by **45%** through systematic diagnostics and post-incident documentation.
- Wrote comprehensive **documentation and educational content** for distributed system architecture, API contracts, and operational runbooks; adapted technical guides based on product updates and team feedback, improving onboarding velocity for **8+ new engineers** and reducing system integration errors by **30%**.
- Contributed to **large-scale infrastructure and networking** projects managing data flows across distributed databases (**PostgreSQL, MySQL, Oracle**) and **AWS storage solutions**; designed ETL pipelines and data validation workflows using **Python and SQL** to ensure data consistency across **50M+ daily transactions** with **99.9% accuracy**.

SKILLS

Software Development & Languages – C++, Java, Python, SQL, REST APIs.

Distributed Systems & Infrastructure – Large-scale system design, data structures, algorithms, compute architecture, storage design, network architecture, AWS (EC2, S3, Lambda, CloudWatch), Docker containerization.

Code Quality & Development Practices – Code reviews, best practices, style guidelines, testability, efficiency optimization, Git, Jenkins CI/CD, change-control management.

Monitoring & Reliability – Splunk, Dynatrace, CloudWatch, SLA management, incident response, debugging, root cause analysis, mean time to resolution (MTTR) optimization.

Data Management & Processing – PostgreSQL, MySQL, Oracle, data structures, ETL pipelines, data validation, data consistency.

Frameworks & Platforms – Spring Boot, Linux, Jira, Confluence, Agile/Scrum.

EDUCATION AND TRAINING

PROJECTS

Distributed Network Transaction Processing Platform (HCL Technologies)

- Architected and developed a **high-throughput distributed network system** processing **50M+ daily transactions** across geographically dispersed **compute nodes** using **Java Spring Boot microservices** and **AWS infrastructure** (EC2 auto-scaling, S3 event-driven architecture, Lambda workers), reducing latency by **38%** compared to monolithic baseline.
- Implemented efficient **data structures and algorithms** for transaction routing and load balancing across the distributed network, including consistent hashing for cache locality and priority queue optimization for SLA-critical requests, improving throughput by **2.3x** and reducing tail latency (p99) from **850ms to 220ms**.
- Designed **monitoring and observability infrastructure** using **Splunk and CloudWatch** with custom dashboards tracking distributed system health across **12+ regions**; enabled real-time alerting on network partitions and cascading failures, reducing undetected incidents by **92%** and achieving **99.95% availability SLA**.
- Led **design reviews** with cross-functional teams evaluating trade-offs between **compute technologies** (container orchestration vs. serverless), **storage architecture** (relational vs. NoSQL), and **network protocols**; documented architectural decisions and adapted guidance based on production learnings, enabling **3 additional teams** to adopt the platform pattern.
- Established **code review standards and best practices** for distributed system development; trained **6 junior engineers** on debugging distributed failures, network tracing, and performance profiling, reducing critical bugs in production by **56%** and establishing a center of excellence for infrastructure code quality.

Real-Time Payment Network Monitoring & Incident Triage System (HCL Technologies)

- Developed **product code in Java Spring Boot and Python** for a real-time monitoring dashboard supporting **distributed payment network operations**, processing **10M+ daily transactions** with **sub-second latency** using **AWS Lambda event processors** and **CloudWatch Logs** for centralized visibility.
- Implemented **large-scale data processing infrastructure** using **Python ETL pipelines** and **SQL queries** across distributed **PostgreSQL and Oracle databases** to correlate network events and detect anomalies, reducing **P1 incident detection time** by **65%** from **8 minutes to 2.8 minutes average**.
- Established automated **debugging and root cause analysis** workflows that traced issues across hardware, network layers, and service operations; documented findings in **Confluence runbooks** adapted from operational feedback, enabling on-call engineers to resolve **83% of incidents** without escalation within **15 minutes**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)